

```
In [42]: # Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

```
In [43]: # 1. Read the CSV.
df = pd.read_csv('Delhi metro.csv')
```

```
In [44]: # 2. Print shape, column names, and dtypes.
df.shape
```

```
Out[44]: (285, 8)
```

```
In [45]: list(df.columns)
```

```
Out[45]: ['ID (Station ID)',
          'Station Names',
          'Dist. From First Station(km)',
          'Metro Line',
          'Opened(Year)',
          'Layout',
          'Latitude',
          'Longitude']
```

```
In [46]: df.dtypes
```

```
Out[46]: ID (Station ID)          int64
          Station Names          str
          Dist. From First Station(km)  float64
          Metro Line              str
          Opened(Year)            str
          Layout                  str
          Latitude                 float64
          Longitude                float64
          dtype: object
```

```
In [47]: # 3. Show the first 10 rows. Which columns are numeric, and which are text?
df.head(10)
```

```
Out[47]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th {
text-align: right; } </style> <thead> <th></th> <th>ID (Station ID)</th> <th>Station Names</th> <th>Dist. From First Station(km)</th>
<th>Metro Line</th> <th>Opened(Year)</th> <th>Layout</th> <th>Latitude</th> <th>Longitude</th> </thead> <tbody> <th>0</th>
<th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> </tbody> <tbody>
</tbody>
```

1	Shaheed Sthal(First Station)	0.0	Red line	08-03-2019	Elevated	28.670611	77.415582
2	Hindon River	1.0	Red line	08-03-2019	Elevated	28.878965	77.415483
3	Arthala	2.5	Red line	08-03-2019	Elevated	28.676999	77.391892
4	Mohan Nagar	3.2	Red line	08-03-2019	Elevated	28.606319	77.106082
5	Shyam park	4.5	Red line	08-03-2019	Elevated	28.698807	28.698807
6	Major Mohit Sharma	5.7	Red line	08-03-2019	Elevated	28.677611	77.358143
7	Raj Bagh	6.9	Red line	08-03-2019	Elevated	28.640860	77.209500
8	Shaheed Nagar	8.2	Red line	08-03-2019	Elevated	28.530780	77.212057
9	Dilshad Garden	9.4	Red line	04-06-2008	Elevated	28.675920	77.321420
10	Jhil Mil	10.3	Red line	04-06-2008	Elevated	28.675790	77.312390

```
In [48]: # 4. Check for missing values across all columns.
df.isna().sum()
```

```
Out[48]: ID (Station ID)          0
         Station Names           0
         Dist. From First Station(km) 0
         Metro Line              0
         Opened(Year)            0
         Layout                  0
         Latitude                0
         Longitude               0
         dtype: int64
```

```
In [49]: # 5.    How many unique metro lines are there? List them with their station counts
         len(df['Metro Line'].unique())
```

```
Out[49]: 13
```

```
In [50]: df['Metro Line'].value_counts()
```

```
Out[50]: Metro Line
         Blue line          49
         Pink line         38
         Yellow line       37
         Voilet line       34
         Red line          29
         Magenta line      25
         Green line        21
         Aqua line         21
         Rapid Metro       11
         Blue line branch   8
         Orange line        6
         Green line branch  3
         Gray line          3
         Name: count, dtype: int64
```

```
In [51]: # 6. Opened (Year) is stored as text like 08-03-2019 – convert it to a proper datetime column.
         df['Opened(Year)'] = pd.to_datetime(df['Opened(Year)'],dayfirst=True)
```

```
In [52]: df.dtypes
```

```
Out[52]: ID (Station ID)          int64
         Station Names           str
         Dist. From First Station(km) float64
         Metro Line              str
         Opened(Year)            datetime64[us]
         Layout                  str
         Latitude                float64
         Longitude               float64
         dtype: object
```

```
In [53]: # 7. ID (Station ID) Looks Like it should be a unique key. Check df ['ID (Station ID)']. duplicated (). sum ()
df['ID (Station ID)'].duplicated().sum()
```

```
Out[53]: np.int64(236)
```

```
In [54]: df['ID (Station ID)'].duplicated().value_counts()
```

```
Out[54]: ID (Station ID)
True      236
False     49
Name: count, dtype: int64
```

```
In [55]: # 8. Are there any duplicate Station Names? Look them up – are they genuinely the same station or two different stations that
df_dup = df[df['Station Names'].duplicated(keep= 'first')]
df_dup
```

```
Out[55]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th {
text-align: right; } </style> <thead> <th></th> <th>ID (Station ID)</th> <th>Station Names</th> <th>Dist. From First Station(km)</th>
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<th>127</th> </tbody> <tbody></tbody>
```

```
1    Yamuna Bank    0.0    Blue line branch    2009-05-10    At-Grade    28.62331    77.26792
```

```
2    Ashok Park Main    1.4          Green line    2010-04-02    Elevated    28.67153    77.15527
```

```
In [56]: df_dup = df[df['Station Names'].duplicated(keep= 'last')]
df_dup
```

```
Out[56]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th {
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<th>123</th> </tbody> <tbody></tbody>
```

34	Yamuna Bank	35.2	Blue line	2009-05-10	At-Grade	28.62331	77.26792
1	Ashok Park Main	0.0	Green line branch	2010-04-02	Elevated	28.67292	77.14420

```
In [57]: # 9. Show all stations on the Blue line.
df9 = df[df['Metro Line'] == 'Blue line']
df9['Station Names']
```

```
Out[57]: 66      Dwarka Sector 21(First station) [Conn: Orange]
        67              Dwarka Sector 8
        68              Dwarka Sector 9
        69              Dwarka Sector 10
        70              Dwarka Sector 11
        71              Dwarka Sector 12
        72              Dwarka Sector 13
        73              Dwarka Sector 14
        74      Dwarka [Conn: Gray]
        75              Dwarka Mor
        76              Nawada
        77              Uttam Nagar West
        78              Uttam Nagar East
        79      Janak Puri West [Conn: Magenta]
        80              Janak Puri East
        81              Tilak Nagar
        82              Subhash Nagar
        83              Tagore Garden
        84      Rajouri Garden [Conn: Pink]
        85              Ramesh Nagar
        86              Moti Nagar
        87      Kirti Nagar [Conn: Green]
        88              Shadipur
        89              Patel Nagar
        90              Rajendra Place
        91              Karol Bagh
        92              Jhandewalan
        93              R K Ashram Marg
        94      Rajiv Chowk [Conn: Yellow]
        95              Barakhamba
        96      Mandi House [Conn: Violet]
        97      Supreme Court (Pragati Maidan)
        98              Indraprastha
        99              Yamuna Bank
       100              Akshardham
       101      Mayur Vihar Phase-1 [Conn: Pink]
       102              Mayur Vihar Extention
       103              New Ashok Nagar
       104              Noida Sector 15
       105              Noida Sector 16
```

```
106                      Noida Sector 18
107      Botanical Garden [Conn: Magenta]
108                      Golf Course
109                      Noida City Center
110                      Noida Sector 34
111      Noida Sector 52 [Conn: Aqua]
112                      Noida Sector 61
113                      Noida Sector 59
114                      Noida Sector 62
Name: Station Names, dtype: str
```

```
In [58]: # 10. Which stations are Underground vs Elevated vs At-Grade? Count each.
df_under = df[df['Layout'] == 'Underground']
len(df_under)
```

```
Out[58]: 68
```

```
In [59]: df_elev = df[df['Layout'] == 'Elevated']
len(df_elev)
```

```
Out[59]: 214
```

```
In [60]: df_grad = df[df['Layout'] == 'At-Grade']
len(df_grad)
```

```
Out[60]: 3
```

```
In [61]: # 11. Find every station more than 40 km from its line's first station.
df11 = df[df['Dist. From First Station(km)']>40]
df11
```

```
Out[61]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th {
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<th>62</th> <th>63</th> <th>64</th> <th>65</th> <th>103</th> <th>104</th> <th>105</th> <th>106</th> <th>107</th>
<th>108</th> <th>109</th> <th>110</th> <th>111</th> <th>112</th> <th>113</th> <th>114</th> <th>189</th> <th>190</th>
<th>191</th> <th>243</th> <th>244</th> <th>245</th> <th>246</th> <th>247</th> <th>248</th> <th>249</th> <th>250</th>
<th>251</th> <th>252</th> <th>253</th> <th>254</th> </tbody> <tbody></tbody>
```

33	Guru Dronacharya	40.9	Yellow line	2010-06-21	Elevated	28.482030	77.102320
34	Sikandarpur [Conn: Rapid]	41.9	Yellow line	2010-06-21	Elevated	28.481820	77.092350
35	MG Road	43.1	Yellow line	2010-06-21	Elevated	28.479570	77.080060
36	IFFCO Chowk	44.2	Yellow line	2010-06-21	Elevated	28.472090	77.071750
37	Huda City Centre	45.7	Yellow line	2010-06-21	Elevated	28.459270	77.072680
38	New Ashok Nagar	40.4	Blue line	2009-11-12	Elevated	28.589160	77.302040
39	Noida Sector 15	41.4	Blue line	2009-11-12	Elevated	28.585120	77.311390
40	Noida Sector 16	42.5	Blue line	2009-11-12	Elevated	28.578190	77.317570
41	Noida Sector 18	43.6	Blue line	2009-11-12	Elevated	28.570810	77.326120
42	Botanical Garden [Conn: Magenta]	44.7	Blue line	2009-11-12	Elevated	28.564090	77.334200
43	Golf Course	45.9	Blue line	2009-11-12	Elevated	28.567140	77.345980
44	Noida City Center	47.2	Blue line	2009-11-12	Elevated	28.574660	77.356080
45	Noida Sector 34	48.1	Blue line	2019-03-09	Elevated	28.480863	77.084888
46	Noida Sector 52 [Conn: Aqua]	49.3	Blue line	2019-03-09	Elevated	28.480863	77.084888
47	Noida Sector 61	50.5	Blue line	2019-03-09	Elevated	28.480863	77.084888
48	Noida Sector 59	51.5	Blue line	2019-03-09	Elevated	28.480863	77.084888
49	Noida Sector 62	52.7	Blue line	2019-03-09	Elevated	28.480863	77.084888
32	Escorts Mujesar	40.1	Voilet line	2015-09-06	Elevated	28.370234	77.314920

33	Sant Surdas - Sihi	41.8	Voilet line	2018-11-19	Elevated	28.354651	77.316226
34	Raja Nahar Singh	43.5	Voilet line	2018-11-19	Elevated	28.340019	77.316428
27	IP Extension	40.3	Pink line	2018-10-31	Elevated	28.628899	77.310198
28	Anand Vihar [Conn: Blue]	41.9	Pink line	2018-10-31	Elevated	28.646753	77.318004
29	Karkar Duma [Conn: Blue]	42.9	Pink line	2018-10-31	Elevated	28.649162	77.306208
30	Karkarduma Court	44.0	Pink line	2018-10-31	Elevated	28.653600	77.295788
31	Krishna Nagar	44.7	Pink line	2018-10-31	Elevated	28.657846	77.290185
32	East Azad Nagar	45.7	Pink line	2018-10-31	Elevated	28.664696	77.284881
33	Welcome [Conn: Red]	46.8	Pink line	2018-10-31	Elevated	28.671800	77.277560
34	Jaffrabad	48.0	Pink line	2018-10-31	Elevated	28.682682	77.274805
35	Maujpur	49.1	Pink line	2018-10-31	Elevated	28.691978	77.279624
36	Gokulpuri	50.4	Pink line	2018-10-31	Elevated	28.702475	77.286125
37	Johri Enclave	51.7	Pink line	2018-10-31	Elevated	28.712880	77.286125
38	Shiv Vihar	52.6	Pink line	2018-10-31	Elevated	28.617538	77.035430

```
In [62]: df11.value_counts(df['Metro Line'])
```

```
Out[62]: Metro Line
Blue line      12
Pink line      12
Yellow line     5
Voilet line     3
Name: count, dtype: int64
```

```
In [63]: # 12. Sort stations by Opened (Year) – which was the very first station to open, and which is the most recent?
df12 = df.sort_values(by='Opened(Year)')
df12.head(1)
```

```
Out[63]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th {
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<th>Metro Line</th> <th>Opened(Year)</th> <th>Layout</th> <th>Latitude</th> <th>Longitude</th> </thead> <tbody> <th>11</th>
</tbody> <tbody></tbody>
```

12	Shahdara	12.5	Red line	2002-12-24	Elevated	28.67345	77.28962
----	----------	------	----------	------------	----------	----------	----------

```
In [64]: # which is the most recent
df12 = df.sort_values(by='Opened(Year)',ascending=False)
df12.head(1)
```

```
Out[64]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th {
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</tbody> <tbody></tbody>
```

2	Nangli	1.5	Gray line	2019-10-04	Elevated	28.6173	77.010437
---	--------	-----	-----------	------------	----------	---------	-----------

```
In [65]: # 13. Average distance from first station, grouped by Metro Line – which line is Longest end-to-end?
df.groupby('Metro Line')['Dist. From First Station(km)'].mean().sort_values(ascending=False)
```

```
Out[65]: Metro Line
Pink line          28.773684
Blue line          26.144898
Yellow line        21.462162
Voilet line        20.617647
Magenta line       17.656000
Red line           16.558621
Aqua line          13.352381
Green line         11.380952
Orange line        10.566667
Rapid Metro        5.709091
Blue line branch   4.000000
Gray line          1.800000
Green line branch  1.066667
Name: Dist. From First Station(km), dtype: float64
```

```
In [66]: # 14. Count of stations per Metro Line, sorted descending.  
df.groupby('Metro Line')['Station Names'].count().sort_values(ascending=False)
```

```
Out[66]: Metro Line  
Blue line          49  
Pink line          38  
Yellow line        37  
Voilet line        34  
Red line           29  
Magenta line       25  
Aqua line          21  
Green line         21  
Rapid Metro        11  
Blue line branch   8  
Orange line        6  
Gray line          3  
Green line branch  3  
Name: Station Names, dtype: int64
```

```
In [67]: # 15. Group by Layout and Metro Line together – which lines are mostly underground vs mostly elevated?  
pd.crosstab(df['Metro Line'],df['Layout'])
```

```
Out[67]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th {
text-align: right; } </style> <thead> <th>Layout</th> <th>At-Grade</th> <th>Elevated</th> <th>Underground</th> <th>Metro
Line</th> <th></th> <th></th> <th></th> </thead> <tbody> <th>Aqua line</th> <th>Blue line</th> <th>Blue line branch</th>
<th>Gray line</th> <th>Green line</th> <th>Green line branch</th> <th>Magenta line</th> <th>Orange line</th> <th>Pink line</th>
<th>Rapid Metro</th> <th>Red line</th> <th>Voilet line</th> <th>Yellow line</th> </tbody> <tbody></tbody>
```

0	21	0
1	44	4
1	7	0
0	2	1
0	21	0
1	2	0
0	10	15
0	1	5
0	26	12
0	11	0
0	29	0
0	23	11
0	17	20

```
In [68]: # 16. How many stations opened each year? Which year had the most openings?
df['Year'] = df['Opened(Year)'].dt.year
df.groupby('Year')['Station Names'].count().sort_values(ascending=False)
```

```
Out[68]: Year
        2018    64
        2010    54
        2019    37
        2005    28
        2017    18
        2009    17
        2011    13
        2015    13
        2004    11
        2006     9
        2002     6
        2013     5
        2003     4
        2008     3
        2014     3
Name: Station Names, dtype: int64
```

```
In [69]: # 17. Extract just the year from Opened (Year) into a new integer column.
df['Year'] = df['Opened(Year)'].dt.year
df
```

```
Out[69]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th {
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<tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> <th>...</th> <th>280</th> <th>281</th> <th>282</th>
<th>283</th> <th>284</th> </tbody> <tbody></tbody>
```

1	Shaheed Sthal(First Station)	0.0	Red line	2019-03-08	Elevated	28.670611	77.415582	2019
2	Hindon River	1.0	Red line	2019-03-08	Elevated	28.878965	77.415483	2019
3	Arthala	2.5	Red line	2019-03-08	Elevated	28.676999	77.391892	2019
4	Mohan Nagar	3.2	Red line	2019-03-08	Elevated	28.606319	77.106082	2019
5	Shyam park	4.5	Red line	2019-03-08	Elevated	28.698807	28.698807	2019
...	...	...	...	...	...	...	...	...
2	Shivaji Stadium	1.9	Orange line	2011-02-23	Underground	28.629010	77.211900	2011
3	Dhaura Kuan [Conn: Pink]	8.3	Orange line	2011-08-15	Elevated	28.591780	77.161550	2011
4	Delhi Aerocity	14.5	Orange line	2011-08-15	Underground	28.548810	77.120920	2011
5	IGI Airport	17.9	Orange line	2011-02-23	Underground	28.556930	77.086690	2011
6	Dwarka Sector 21 [Conn: Blue]	20.8	Orange line	2011-02-23	Underground	28.552260	77.058280	2011

<p>285 rows × 9 columns</p>

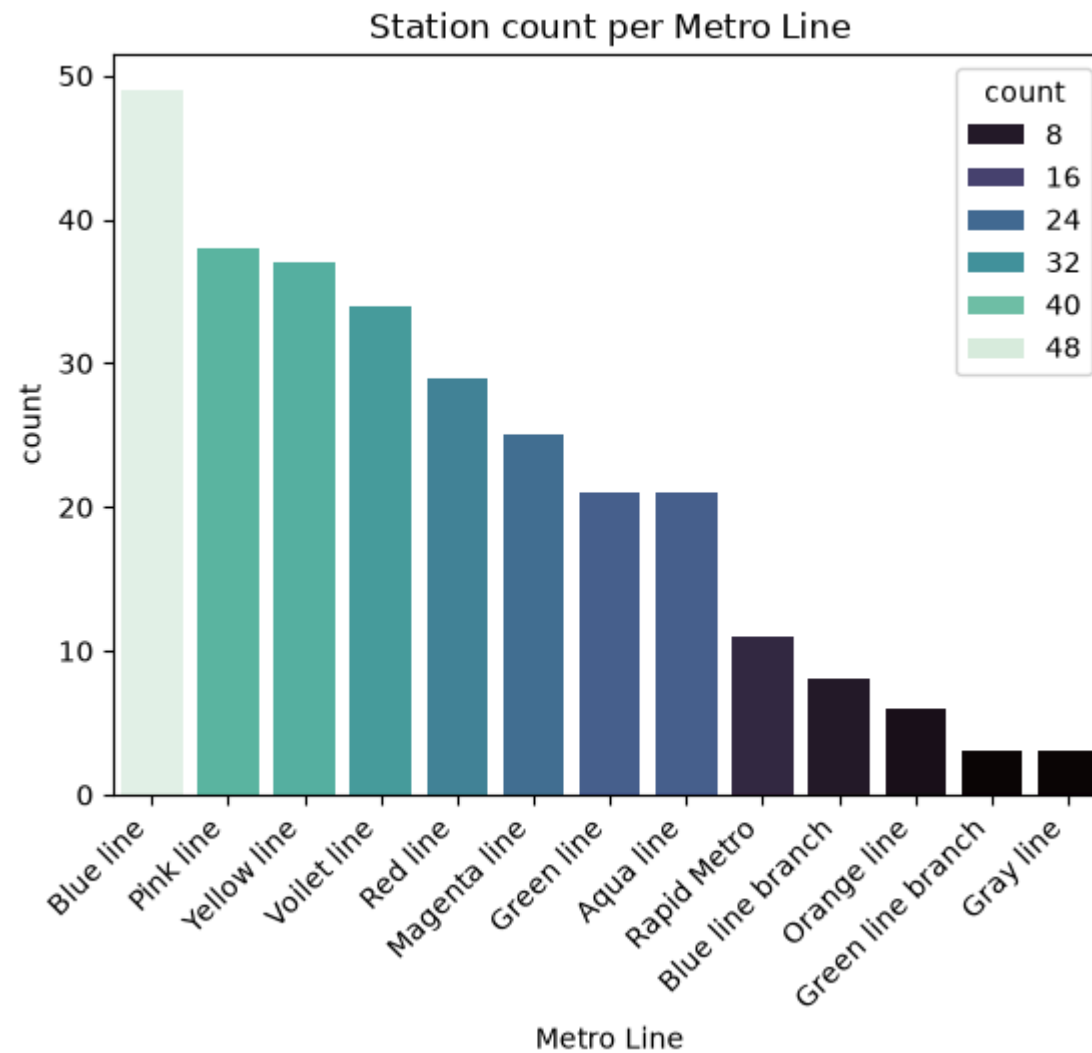
```
In [70]: # 18. Bar plot of station count per Metro Line
df18 = df['Metro Line'].value_counts().reset_index()
df18
```

```
Out[70]: <style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th {
text-align: right; } </style> <thead> <th></th> <th>Metro Line</th> <th>count</th> </thead> <tbody> <th>0</th> <th>1</th>
<th>2</th> <th>3</th> <th>4</th> <th>5</th> <th>6</th> <th>7</th> <th>8</th> <th>9</th> <th>10</th> <th>11</th>
<th>12</th> </tbody> <tbody></tbody>
```

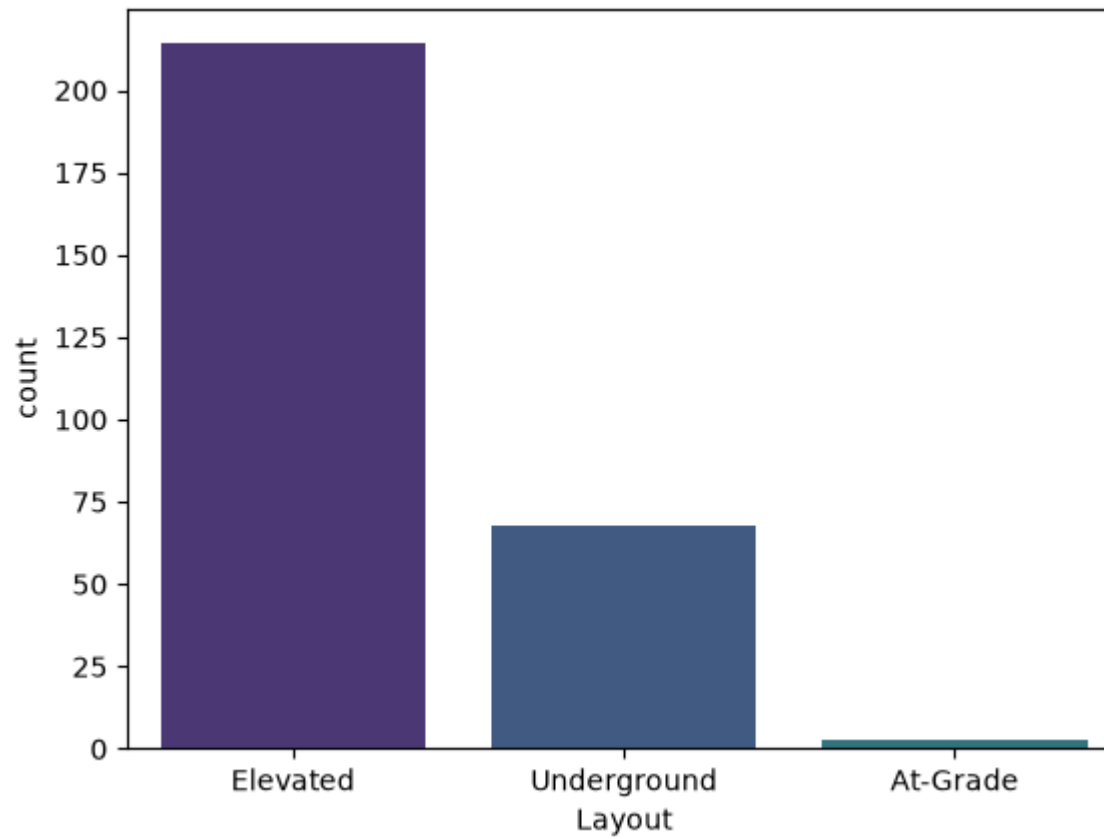
Blue line	49
Pink line	38
Yellow line	37
Voilet line	34
Red line	29
Magenta line	25
Green line	21
Aqua line	21
Rapid Metro	11
Blue line branch	8
Orange line	6
Green line branch	3
Gray line	3

```
In [71]: # plot
plt.figure(figsize=(7,5))
sns.barplot(data=df18,x='Metro Line',y='count',hue='count',palette=sns.color_palette("mako", as_cmap=True))
plt.title('Station count per Metro Line')
plt.xticks(rotation = 45,ha = 'right')
```

```
Out[71]: ([0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12],
          [Text(0, 0, 'Blue line'),
           Text(1, 0, 'Pink line'),
           Text(2, 0, 'Yellow line'),
           Text(3, 0, 'Voilet line'),
           Text(4, 0, 'Red line'),
           Text(5, 0, 'Magenta line'),
           Text(6, 0, 'Green line'),
           Text(7, 0, 'Aqua line'),
           Text(8, 0, 'Rapid Metro'),
           Text(9, 0, 'Blue line branch'),
           Text(10, 0, 'Orange line'),
           Text(11, 0, 'Green line branch'),
           Text(12, 0, 'Gray line')])
```



```
In [72]: # 19. Countplot of Layout (Elevated/Underground/At-Grade).  
plt.figure(figsize=(7,5))  
cp = sns.countplot(data=df, x='Layout', hue='Layout', order=df['Layout'].value_counts().index, palette=sns.color_palette('viridis'))
```



```
In [73]: # 20. Histogram of Dist. From First Station(km) – what shape is the distribution?
sns.histplot(data=df,x='Dist. From First Station(km)',bins=25,kde=True)
```

```
Out[73]: <Axes: xlabel='Dist. From First Station(km)', ylabel='Count'>
```

